

Vertex algebras with nice structure despite failing all conventional niceness criteria

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Joint work with Robert Allen and David Ridout

Outline

- ① Conformal symmetry and conformal field theory
- ② Vertex algebras
- ③ Examples

Conformal symmetry

Definition

A *conformal* transformation is a transformation that preserves angles.

Proposition

In 2 dimensions there are infinitely many independent (local) conformal transformations, because any meromorphic function is conformal away from the zeros of its derivative.

The central extension of the Lie algebra of local conformal transformations is called the *Virasoro algebra*.

Basis: $c, L_n, n \in \mathbb{Z}$

Relations: $[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}$, $[L_n, c] = 0$.

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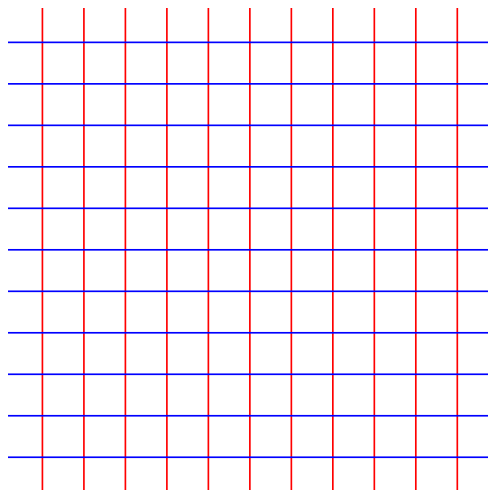
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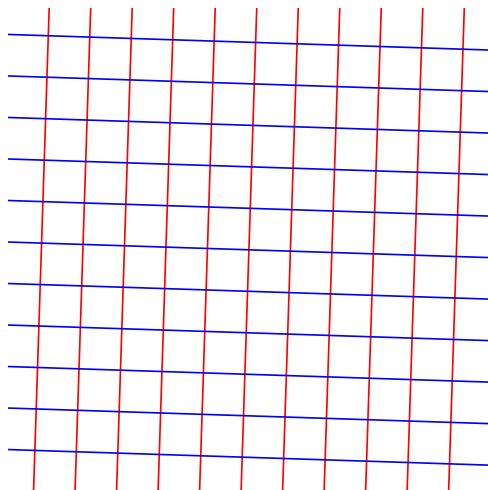
Examples of conformal transformations

Translation



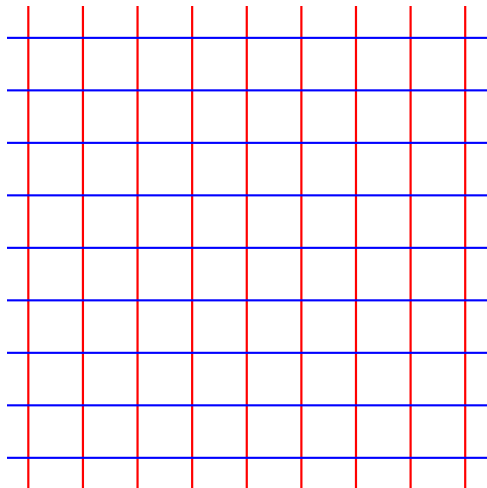
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Rotation



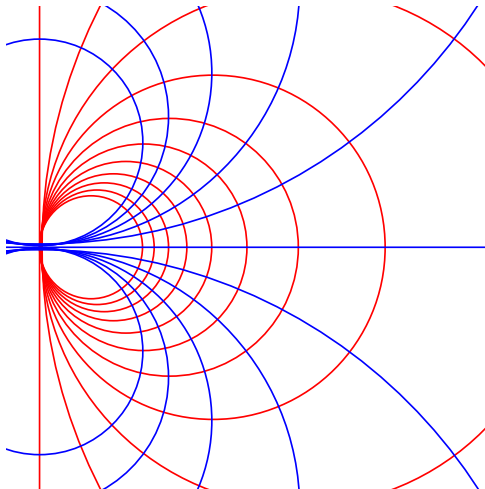
Examples of conformal transformations

Dilation (zoom in or out)



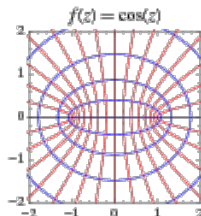
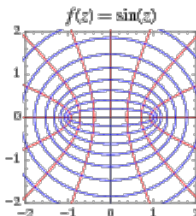
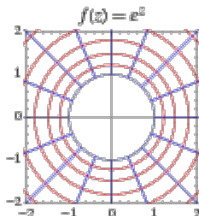
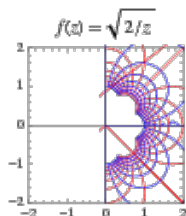
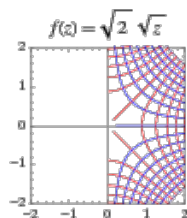
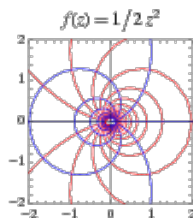
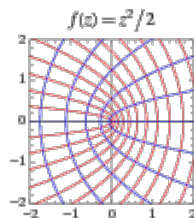
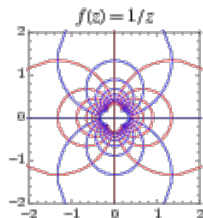
Examples of conformal transformations

Special conformal transformation



Examples of conformal transformations

Local conformal transformations specific to 2D



Real world examples of conformal symmetry

- Statistical physics models such as the Ising model
- Quantum field theory at very high energies
- String theory

Vertex algebras

Vertex algebras encode conformal symmetry in 2D quantum field theories.

Definition: vertex algebra

Data:

- graded vector space
 $V = \sum_{n=0}^{\infty} V[n],$
- vacuum vector $|0\rangle \in V[0],$
- conformal vector $\omega \in V[2],$
- field maps
 $Y(\cdot, z) : V \rightarrow \text{End}(V)[[z, z^{-1}]]$

Axioms:

- $Y(|0\rangle, z) = \text{Id}_V.$ For all $A \in V$
 $Y(A, z) |0\rangle = A + V[[z]]z$
- $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$
- $L_0|_{V[n]} = n \cdot \text{id}$
- For all $A \in V$
 $[T, Y(A, z)] = Y(L_{-1}A, z) = \partial Y(A, z)$
- For every $A, B \in V,$ there exists
 $N \in \mathbb{N}$ such that
 $(z-w)^N [Y(A, z), Y(B, w)] = 0.$

Some features of rational conformal field theories

The nicest class of vertex algebras is called *rational* (definition on next slide).

- Space of states is a module over 2 commuting copies of a rational vertex algebra (chiral algebra).
- Singularities of correlation functions are at worst rational poles.
E.g.

$$\frac{1}{|z-w|^{\frac{1}{4}}}$$

Correlation functions define a kind of tensor product of modules and turn module category into a tensor category.

- Character of space of states is a modular invariant.

Rational vertex algebras

- Module theory is semisimple.
→ Virasoro L_0 operator is diagonalisable.
- Finitely many inequivalent simple modules.
- Span of module characters carries a representation of the modular group $SL(2, \mathbb{Z})$. [Zhu]
- Module category is a *modular tensor category*. [Verlinde, Moore-Seiberg, Huang]

Fusion product derived
from correlation functions

=

Verlinde product derived
from $SL(2, \mathbb{Z})$ action on
characters

Features of logarithmic conformal field theories

- Space of states is a module over 2 commuting copies of a **logarithmic** vertex algebra.
- Singularities of correlation functions can be **logarithmic**. *E.g.*

$$\log|z - w|$$

Correlation functions still define a tensor product. **Product of simple modules need not be semisimple.**

- Character of space of states is still a modular invariant.

Logarithmic vertex algebras

- Module theory is **not** semisimple.
→ Virasoro L_0 operator can have **Jordan blocks**.
- Finitely many inequivalent simple modules. **May or may not fail**.
- Span of **torus 1-point functions** carries a representation of the modular group $SL(2, \mathbb{Z})$, if there are only finitely many simple modules. [Miyamoto]
- Module category is **not** a *modular tensor category*.
Verlinde formula must fail as characters can't distinguish all modules and modular action does not close on characters.

Examples of logarithmic conformal field theories

The symplectic fermions are generated by two odd fields ξ_1, ξ_2 :

$$\xi_1(z)\xi_2(w) \sim \frac{1}{(z-w)^2} \sim -\xi_2(z)\xi_1(w), \quad \xi_1(z)\xi_1(w) \sim 0 \sim \xi_2(z)\xi_2(w).$$

The even subalgebra is called “the $c = -2$ triplet”.

Defines a logarithmic vertex algebra/conformal field theory.

[Gaberdiel-Kausch]

$c = -2$ triplet module theory

4 simple modules:

$$S_0, \quad S_1, \quad S_{\frac{-1}{8}}, \quad S_{\frac{3}{8}}$$

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Indices denote conformal highest weight.

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Vertex algebra.

$c = -2$ triplet module theory

4 simple modules:

$$S_0, \quad S_1, \quad S_{-\frac{1}{8}}, \quad S_{\frac{3}{8}}$$

Cannot be extended to form reducible yet indecomposable modules (projective and injective).

$c = -2$ triplet module theory

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Can be extended to form reducible yet indecomposable modules.

Simple modules are not closed under fusion

S_0 fusion unit

$$S_1: S_0 \leftrightarrow S_1, S_{\frac{-1}{8}} \leftrightarrow S_{\frac{3}{8}}$$

$$S_{\frac{-1}{8}} \times S_{\frac{-1}{8}} = S_{\frac{3}{8}} \times S_{\frac{3}{8}} = P_0, S_{\frac{-1}{8}} \times S_{\frac{3}{8}}$$

$$S_1: P_0 \leftrightarrow P_1$$

$$S_i \times P_j = 2S_{\frac{-1}{8}} \oplus 2S_{\frac{3}{8}}, i = \frac{-1}{8}, \frac{3}{8}, j = 0, 1$$

Theorem [Tsuchiya-Kanie]

The new modules P_0, P_1 are indecomposable yet reducible, but cannot be further extended (projective and injective). All indecomposables classified.

Theorem [Tsuchiya-SW]

Above fusion formulae hold. In particular, fusion closes on $S_0, S_1, S_{\frac{-1}{8}}, S_{\frac{3}{8}}, P_0, P_1$. Module category is a rigid tensor category.

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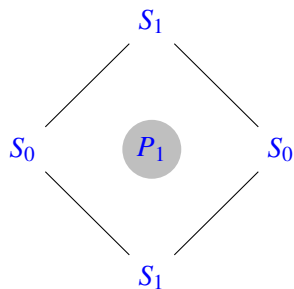
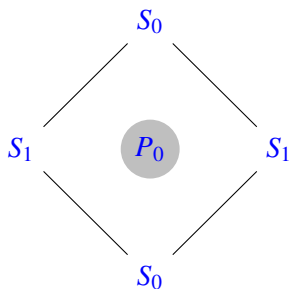
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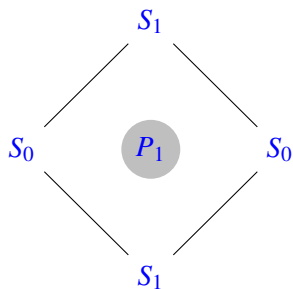
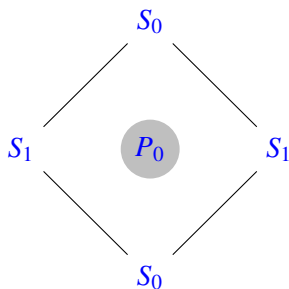
Submodule structure of P_0 and P_1



Module characters: $\text{ch}[M](q) = \text{tr}_M q^{L_0 - \frac{c}{24}}$, $q = e^{2\pi i \tau}$, here $c = -2$.

Characters cannot distinguish indecomposables from the sum of their composition factors: $\text{ch}[P_0] = \text{ch}[P_1] = 2\text{ch}[S_0] + 2\text{ch}[S_1]$

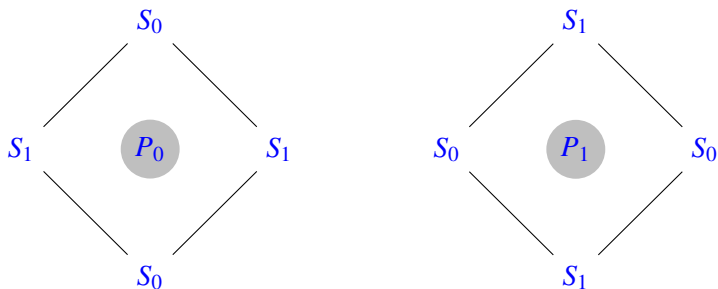
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The Verlinde formula for rational vertex algebras

Modular group: $\mathrm{SL}(2, \mathbb{Z}) = \langle S, T \mid S^2 = (ST)^3, S^4 = 1 \rangle$
 $T : \tau \mapsto \tau + 1, S : \tau \mapsto -\frac{1}{\tau}$

Action of $\mathrm{SL}(2, \mathbb{Z})$ closes on span of rational vertex algebra module characters.

Expand in basis of simple module characters.

$$T : \mathrm{ch}[M_i] \mapsto T_i \mathrm{ch}[M_i]$$

$$S : \mathrm{ch}[M_i] \mapsto \sum_j S_{i,j} \mathrm{ch}[M_j]$$

The Verlinde formula relates the fusion structure constants and the S -matrix coefficients.

$$M_i \times M_j = \bigoplus_k N_{i,j}^k M_k$$

$$N_{i,j}^k = \sum_n \frac{S_{i,n} S_{j,n} \overline{S_{k,n}}}{S_{0,n}}$$

A logarithmic Verlinde formula?

- Verlinde formula is derived from characters. At best it could predict fusion at the level of characters.
- For logarithmic vertex algebras S -transformations have a polynomial τ dependence. This makes no sense in the Verlinde formula, as it must produce non-negative integers.
- We solve this paradox by giving up on having only finitely many simple modules.
- If done properly the τ -dependence will be restricted to a ‘subset of modules of measure 0’.

The $\beta\gamma$ ghost logarithmic conformal field theory

Two generating fields β and γ .

$$\gamma(z)\beta(w) \sim \frac{1}{z-w} \sim -\beta(z)\gamma(w), \quad \beta(z)\beta(w) \sim 0 \sim \gamma(z)\gamma(w)$$

Conformal field $T(z) = - : \beta(z)\partial\gamma(z) :$, central charge $c = 2$.

Closely related to the Lie algebra spanned by $\mathbf{1}, \beta_n, \gamma_n, n \in \mathbb{Z}$.

$$[\gamma_m, \beta_n] = \delta_{m+n,0}\mathbf{1}, \quad [\beta_m, \beta_n] = 0 = [\gamma_m, \gamma_n]$$

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Highest weight modules (category $\mathcal{O}$)

Category $\mathcal{O}$ (the natural category for highest weight modules) contains only one highest weight module $\mathcal{V}$ and it is simple.

Closed under fusion, $\mathcal{V} \times \mathcal{V} = \mathcal{V}$, but not under action of $\mathrm{SL}(2, \mathbb{Z})$.

Go beyond highest weight modules by considering *parabolic Verma modules*. This module category is no longer semisimple.

Simples: $\mathcal{V}, \mathcal{V}^*, \mathcal{W}_\lambda, \lambda \in \mathbb{R}/\mathbb{Z}, \lambda \neq \mathbb{Z}$.

Not closed under fusion or $\mathrm{SL}(2, \mathbb{Z})$.

Extend again by twisting by an algebra automorphism σ , called *spectral flow*, and allowing for finite length (weight) extensions.

Simples: $\sigma^\ell \mathcal{V}, \sigma^\ell \mathcal{W}_\lambda, \lambda \in \mathbb{R}/\mathbb{Z}, \lambda \neq \mathbb{Z}, \ell \in \mathbb{Z}$.

Topological basis of characters: $\sigma^\ell \mathcal{W}_\lambda, \lambda \in \mathbb{R}/\mathbb{Z}, \ell \in \mathbb{Z}$.

Closes under action of $\mathrm{SL}(2, \mathbb{Z})$. [Ridout-SW]

Closes under fusion. [Allen-SW]

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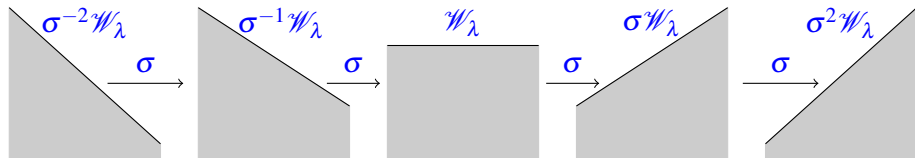
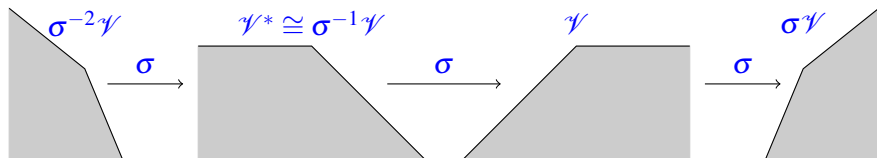
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Spectral flow twists

Spectral flow 'tilts' the energy grading:



Towards a logarithmic Verlinde formula

The action of the modular group closes on the span of $\sigma^{\ell\mathscr{W}}_{\lambda}$ characters.

$$S : \text{ch} \left[\sigma^{\ell\mathscr{W}}_{\lambda} \right] \mapsto \sum_{m \in \mathbb{Z}} \int_{\mathbb{R}/\mathbb{Z}} S \left[\sigma^{\ell\mathscr{W}}_{\lambda} \rightarrow \sigma^{m\mathscr{W}}_{\mu} \right] \text{ch} \left[\sigma^{m\mathscr{W}}_{\mu} \right] d\mu$$
$$S \left[\sigma^{\ell\mathscr{W}}_{\lambda} \rightarrow \sigma^{m\mathscr{W}}_{\mu} \right] = (-1)^{\ell+m} e^{-2\pi i(\ell\mu+m\lambda)}$$

No τ dependence!

What about $\sigma^{\ell\mathscr{V}}$?

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Towards a logarithmic Verlinde formula

We can resolve $\mathcal{V}$ in terms of the $\sigma^\ell \mathcal{W}_0$.

$$\cdots \longrightarrow \sigma^{-2} \mathcal{W}_0 \longrightarrow \sigma^{-1} \mathcal{W}_0 \longrightarrow \mathcal{W}_0 \longrightarrow \mathcal{V} \longrightarrow 0$$

The character of $\mathcal{V}$ can then be computed using the Euler-Poincaré principle.

$$\mathrm{ch} \left[\sigma^\ell \mathcal{V} \right] = \sum_{m=0}^{\infty} (-1)^m \mathrm{ch} \left[\sigma^{\ell-m} \mathcal{W}_0 \right].$$

S-transformation:

$$\mathrm{S} \left[\sigma^\ell \mathcal{V} \rightarrow \sigma^m \mathcal{W}_\mu \right] = (-1)^{\ell+m+1} \frac{e^{-2\pi i(\ell+1/2)\mu}}{e^{\pi i\mu} - e^{-\pi i\mu}}$$

Towards a logarithmic Verlinde formula

We can now propose a Verlinde formula for the characters of fusion products.

$$\mathrm{ch}[M \times N] = \sum_{m \in \mathbb{Z}} \int_{\mathbb{R}/\mathbb{Z}} N_{M,N}^{m,\mu} \mathrm{ch}[\sigma^m \mathscr{W}_\mu] \, \mathrm{d}\mu.$$

The natural generalisation of the rational Verlinde formula is:

$$N_{M,N}^{m,\mu} = \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}/\mathbb{Z}} \frac{\mathrm{S}[M \rightarrow \sigma^n \mathscr{W}_v] \mathrm{S}[N \rightarrow \sigma^n \mathscr{W}_v] \overline{\mathrm{S}[\sigma^m \mathscr{W}_\mu \rightarrow \sigma^n \mathscr{W}_v]}}{\mathrm{S}[\mathscr{V} \rightarrow \sigma^n \mathscr{W}_v]} \, \mathrm{d}v$$

Towards a logarithmic Verlinde formula

Fusion predicted by Verlinde formula:

$$\begin{aligned}\mathrm{ch} \left[\sigma^{\ell} \mathcal{V} \times \sigma^m \mathcal{V} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{V} \right], \\ \mathrm{ch} \left[\sigma^{\ell} \mathcal{V} \times \sigma^m \mathcal{W}_{\mu} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{W}_{\mu} \right], \\ \mathrm{ch} \left[\sigma^{\ell} \mathcal{W}_{\lambda} \times \sigma^m \mathcal{W}_{\mu} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{W}_{\lambda+\mu} \right] + \mathrm{ch} \left[\sigma^{\ell+m-1} \mathcal{W}_{\lambda+\mu} \right].\end{aligned}$$

Fusion products uniquely determined by characters unless $\lambda + \mu \in \mathbb{Z}$.
This matches fusion computed from first principles.

Towards a logarithmic Verlinde formula

Fusion predicted by Verlinde formula:

$$\begin{aligned}\mathrm{ch} \left[\sigma^{\ell} \mathcal{V} \times \sigma^m \mathcal{V} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{V} \right], \\ \mathrm{ch} \left[\sigma^{\ell} \mathcal{V} \times \sigma^m \mathcal{W}_{\mu} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{W}_{\mu} \right], \\ \mathrm{ch} \left[\sigma^{\ell} \mathcal{W}_{\lambda} \times \sigma^m \mathcal{W}_{\mu} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{W}_{\lambda+\mu} \right] + \mathrm{ch} \left[\sigma^{\ell+m-1} \mathcal{W}_{\lambda+\mu} \right].\end{aligned}$$

Fusion products uniquely determined by characters unless $\lambda + \mu \in \mathbb{Z}$.

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Towards a logarithmic Verlinde formula

Fusion predicted by Verlinde formula:

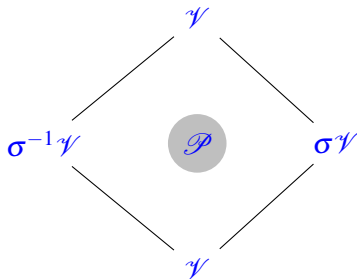
$$\begin{aligned}\mathrm{ch} \left[\sigma^{\ell} \mathcal{V} \times \sigma^m \mathcal{V} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{V} \right], \\ \mathrm{ch} \left[\sigma^{\ell} \mathcal{V} \times \sigma^m \mathcal{W}_{\mu} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{W}_{\mu} \right], \\ \mathrm{ch} \left[\sigma^{\ell} \mathcal{W}_{\lambda} \times \sigma^m \mathcal{W}_{\mu} \right] &= \mathrm{ch} \left[\sigma^{\ell+m} \mathcal{W}_{\lambda+\mu} \right] + \mathrm{ch} \left[\sigma^{\ell+m-1} \mathcal{W}_{\lambda+\mu} \right].\end{aligned}$$

Fusion products uniquely determined by characters unless $\lambda + \mu \in \mathbb{Z}$.

This matches fusion computed from first principles.

Indecomposable fusion products

By direct computation: $\mathscr{W}_\lambda \times \mathscr{W}_{-\lambda} = \sigma^{-1} \mathscr{P}$, where



$$\mathrm{ch} [\mathscr{W}_\lambda \times \mathscr{W}_{-\lambda}] = \mathrm{ch} [\sigma^{-1} \mathscr{P}] = \mathrm{ch} [\mathscr{W}_0] + \mathrm{ch} [\sigma^{-1} \mathscr{W}_0].$$

Conjecture [Ridout-SW], Theorem [Allen-SW]

Let $\mathcal{C}$ be the abelian category of $\beta\gamma$ vertex operator algebra modules generated by the closure under finite length weight extensions of the $\sigma^l \mathcal{V}$ and $\sigma^l \mathcal{W}_\lambda$, $l \in \mathbb{Z}$, $\lambda \in \mathbb{R}/\mathbb{Z}$, $\lambda \neq \mathbb{Z}$. Then,

- the $\sigma^l \mathcal{W}_\lambda$ are simple, projective and injective in $\mathcal{C}$,
- the $\sigma^l \mathcal{P}$ are indecomposable projective covers and injective hulls of $\sigma^l \mathcal{V}$ in $\mathcal{C}$,
- the logarithmic Verlinde formula holds.

Conclusion

- Logarithmic conformal field theories/vertex algebras admit reducible yet indecomposable modules.
- Having finitely many simple modules seems to break the Verlinde formula.
- Verlinde formula can be fixed by allowing infinitely many simple modules.